

ME1103

Principles of Mechanics and Materials

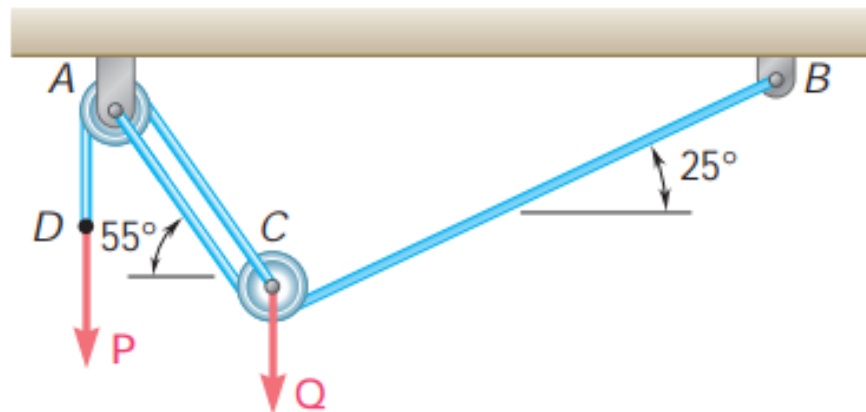
PROBLEM SET 2

Question 1

A 1800-N load Q is applied to the pulley C , which can roll freely on the cable ACB . The pulley is held in the position shown by a second cable CAD , which passes over the pulley A and supports a load P . Determine

- (a) the tension in cable ACB ,
- (b) the magnitude of load P .

Answer: (a) $T_{ACB} = 1048\text{ N}$, and (b) $P = 608\text{ N}$

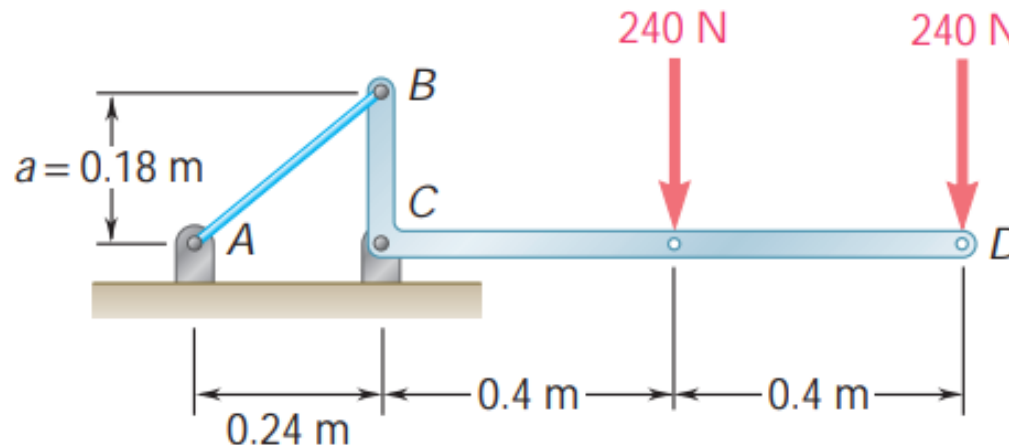


Question 2

The bracket BCD is hinged at C and attached to a control cable at B . For the loading shown, determine:

- (a) the tension in the cable, and
- (b) the reaction at C .

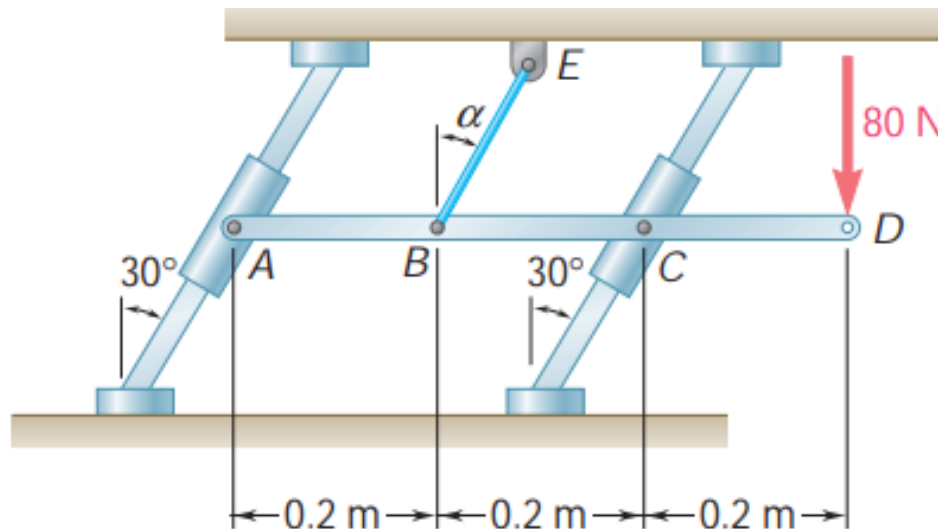
Answer: (a) $T = 2 \text{ kN}$; (b) $C_x = 1.6 \text{ kN}, C_y = 1.68 \text{ kN}$



Question 3

Bar AD is attached at A and C to collars that can move freely on the rods shown. If the cord BE is inclined at $\alpha = 30^\circ$ w.r.t. the vertical, determine the tension in the cord and the reactions at A and C .

Answer: (a) $T = 69.3 \text{ N}$; (b) $A = 140 \text{ N}, C = 180 \text{ N}$



Question 4

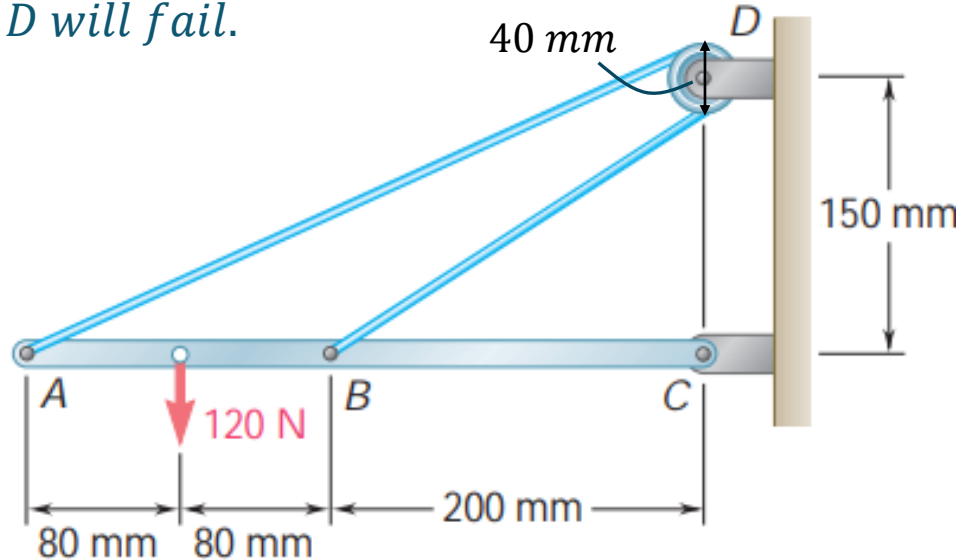
For the assembly shown in the figure, determine:

- The tension in the cable, and
- Suppose the pin joints at C and D can only withstand a maximum load of 250 N . Can they survive this loading condition?

Answer:

(a) $T = 128\text{ N}$

(b) *Pin joint D will fail.*

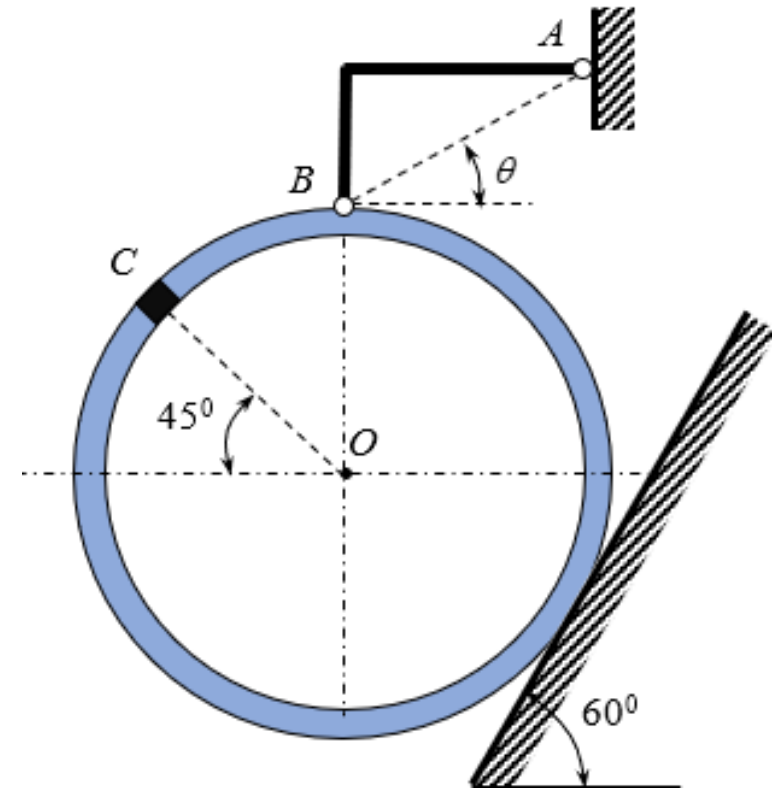


Question 5

A 10 kg ring of negligible thickness and mean diameter of 4 m is resting on a smooth inclined surface and is held in position by an angled bar AB as shown in the figure. The bar is hinged and connected to the ring at B and the wall at A . A point mass C of 3 kg is fixed to the ring at position shown.

Given that point B is vertically above the centre of the ring, determine the angle θ for the assembly to be in static equilibrium. Use $g = 10\text{ m/s}^2$.

Answer: $\theta = 79.8^\circ$



Question 6

For the given supporting structure, determine:

- The magnitude of the reaction forces at pins A and D when the suspended load at hook is 12 kN .
- Suppose the anchoring point of the rope can be shifted along the wall between the brackets. Where should it be positioned w.r.t. bracket at A so that force in member BD is only 24 kN .

Answer:

(a) $F_D = 30\text{ kN}$ and $F_A = 26.8\text{ kN}$

(b) Anchor at about 1.74 m above A .

